# Developing RSE skills at the novice level

Student RSE model for research support, experiential learning, and technical workforce development

**Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan**
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

## Paid students contribute to research and build RSE skills while they do it.




Campuses need more data and software help than staff can provide. Courses leave gaps, and so does the code researchers write. DataSquads are the glue. Many students never hear of this work, especially those from underrepresented groups.

## A paid job that teaches

Students are paid employees, supervised by library or department staff. They help researchers with data from wrangling to reproducibility.

### Who is on a squad

- Student **data analysts and developers** do the hands-on work with data and code.
- A student **project manager** triages requests.
- Student **technical writers** document projects.
- **Staff** supervise, mentor, fund, and sustain the squad.

### Limits we keep

We help with a researcher's data and code. We don't change their research questions, tutor their students, or share what we learn in confidence.

### What slows us down

Student availability, the academic calendar, onboarding, review, and graduation handoffs all factor into the cycle and pace of work. More students take on more varied projects but need more oversight. Handbooks, templates, and staff are what stay when students graduate.

## What the work looks like

CARLETON **Election-news scraper.** Nine students started, extended, or used it from 2021 to 2024 to collect broadcast transcripts for a political science project. **The tool outlasted the students who began it.**

UCLA **wbcheck.** Students built the first version of this Carpentries lesson checker in 2023, and it grew as we learned what collaborative lesson development takes. Staff expanded it in 2026 with AI assistance. It has 48 documented rules, tests, a BSD-3 license, and a citation file crediting its student authors (doi:10.5281/zenodo.23329531).

CARLETON **Language placement dashboard.** The project ran longer than any student would stay, so the squad sketched the database and chose heatmaps for the development team to build.

UCLA **UC OSPO Education site.** Students built the lesson pages and search for a catalog of 55 open source lessons on open science and research software development. It has a citation file and a DOI (doi:10.5281/zenodo.23330022) crediting its developers.

**Research software habits, when the project needs them.** Run instructions, a dependency list, one test, a citation file. We don't treat them as a checklist, and we plan to track whether clients keep them.

## Where students go next

We located 19 UCLA former student employees. By current job category (public profiles, 2026):

| | |
|---|---|
| Data / analytics / consulting | 11 |
| Software / engineering / IT | 4 |
| Research / graduate school | 2 |
| Nonprofit / public sector | 2 |

Descriptive, not a placement rate.

## What adopting takes

- Paid student positions and a wage budget
- Supervisors and mentors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

## Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise student workers? Compare onboarding, review, and graduation handoffs.

**datasquad.info/join**